

Raspberry Pi) to a laptop or computer. In IoT terms, that processing device is called a gateway.

In this particular use case, it is absolutely not required to send the data to the cloud. Instead, it can go to that laptop or computer that has enough power to process it. The entire process can be carried out in the room itself, without third party systems integrators or cloud vendors like AWS, Google and Huawei getting involved.

The advantages of moving to edge computing are listed below:

- Reduces the cost of shipping large volumes of data to the cloud for processing and storage.
- Provides trust and security to the many organisations that have sensitive data, which they don't want to move beyond their premises under any circumstances.
- Enables real-time decision making with ultra-low latency.
- Provides independent and offline connectivity, as intermittent cloud connectivity is a serious concern for mission-critical IoT applications.

The key requirements for edge computing are:

- Maximum computing resources at the edge
- Offline scenarios (i.e., the capability to operate in disconnected mode) and communication
- Security and privacy
- The platform should be lightweight and scalable
- Robust device life cycle
- Data analytics capabilities
- Remote management
- Efficient energy management

To meet the above requirements, you need a powerful software platform besides a bunch of Raspberry Pi/IoT devices.

Open source edge platforms

There is no competition between IoT and edge computing. In the IoT (the Internet of Things), devices are connected to the Internet so that they can communicate with each other. However, it is not just about

connecting to the Internet but gathering intelligence from the data at the edge or near its source. To provide intelligence to things, you need to do some computing. This is where edge computing comes into the picture.

There are many open source edge computing platforms today such as EdgeX Foundry, Akraino Edge Stack, StarlingX, Home Edge, Opnfv, Onap, Kura and more. In addition, many more players are coming up with edge computing platforms. However, this is creating a lot of fragmentation with many such players coming together and doing something of their own.

A unified open source framework for the edge

In January this year, the Linux Foundation announced the launch of LF Edge, an umbrella organisation to establish an open, interoperable framework for edge computing, independent of hardware, silicon, cloud, or operating system. Initially, it included five projects to support emerging edge applications in the area of non-traditional video, and connected things that require lower latency, faster processing and mobility.

LF Edge includes Akraino Edge Stack, EdgeX Foundry and Open Glossary of Edge Computing, formerly standalone projects at The Linux Foundation. The initiative also includes Home Edge Project, a project contributed by Samsung Electronics, and Project EVE (Edge Virtualisation Engine), contributed by ZEDED.

EdgeX Foundry: The architecture of this platform is very systematic and modular. It can run in Raspberry Pi or in any gateway kind of device which has got more compute, like your laptop.

There is no ML algorithm inbuilt in edge computing platforms. On pure edge computing platforms, you can run different applications or business use cases. Data analytics is the key requirement for the edge; so edge computing platforms will give you a data analytics pipeline engine.

KubeEdge: KubeEdge is a CNCF sandbox project, which was originally contributed by Huawei. It is an open source edge computing platform. It is slightly different from other edge computing platforms because it is based on Kubernetes, an open source system for easy management and discovery of applications, and it's fully cloud native.

KubeEdge is simply an extension of the cloud. It is just a matter of pushing the cloud into the edge without complicating things. Even though it's based on Kubernetes, KubeEdge can work independently. It can work in the node without connecting to the cloud.

Roof computing

Roof computing is a standard from IEEE to help drive edge computing standardisation. It is one of the first specifications that is being driven from India.

Amongst the top ten strategic technology trends in 2019, as per Gartner and Forrester, empowered edge is ranked at the fifth position.

Edge computing is not a notion anymore but the reality. We can now see a lot of devices coming up with more computing power. The best example is your smartphone. 

Transcribed by: Dineshwori Longjam, OSFY Bureau

About Authors

The above article is an extract from a talk by Sanil Kumar D. and Krishna Kumar, chief architects at Huawei, at the India Electronics Week - *IoTshow.in*, in February 2019.